

Feb 1, 2022

Introduction to Computer and Network Security

Part 1

Question 1 [40 points]

Explain how to protect a password file with hashing and salting; discuss (a) what is hashing, (b) what is salting, (c) why hashing is not enough for protecting passwords and how salting mitigates the problem of hashing, (d) describe the structure of a password file, and (e) explain if salts can be stored in clear.

Question 2 [10 points]

Define the notions of vulnerability and threat and give (at least) an example for each one.

Part 2

Question 3 [40 points]

In the context of the SAML standard, (a) explain its main goal, (b) describe the two scenarios for the Web Single Sign On profile, namely Identity provider initiated and Service provider initiated, (c) explain how is it possible to guarantee trust in the assertions consumed by the relying party (hint: remember that assertions are cryptographically signed), and (d) explain how it is possible to avoid man-in-the-middle attacks.

Question 4 [10 points]

Explain the notion of pseudo anonymization function, describe at least two possible implementations together with their advantages and disadvantages.

Please, remember...

On each sheet, make sure to write: Name Surname Student ID number Question number	In case of online exam, the file containing the scan of your answers should be named as follows: surname-name-studentIDnumber.pdf
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